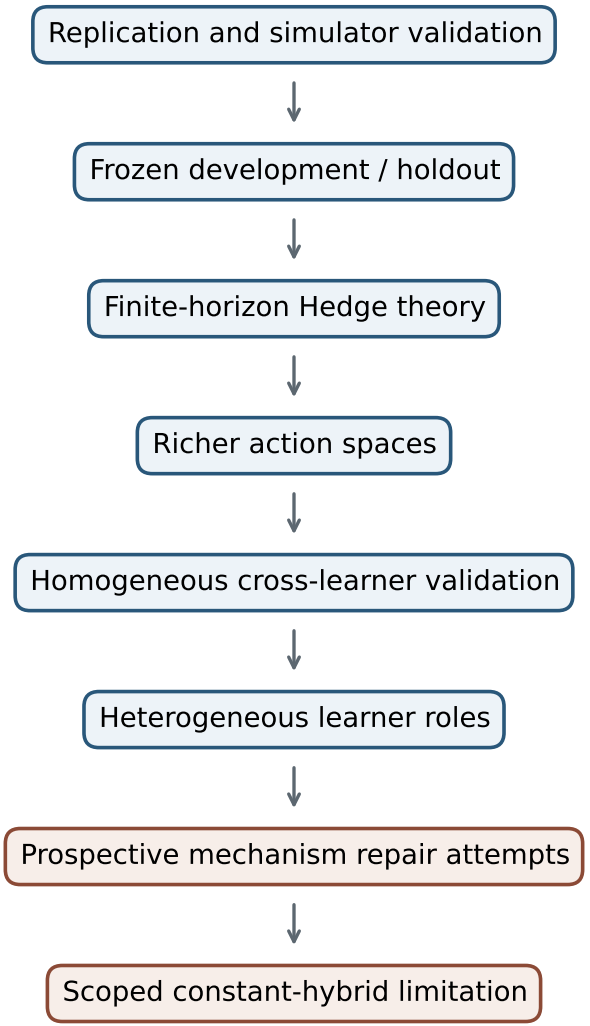Evidence ladder: discovery, generalization, and design limitation